

# Appendix for “Estimating Representative Causal Effects with Double Machine Learning”

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## A Detailed Derivations

### A.1 Derivation of the Conditional Causal Derivative

Note that, in our stylized example,  $T_i$  is  $\text{Poisson}(\lambda_j)$  conditional on  $X_i = j$ . Therefore, its probability mass function is given by:

$$P(T_i = t | X_i = j) = e^{-\lambda_j} \frac{\lambda_j^t}{t!}, t = 0, 1, 2, \dots, \quad (1)$$

and the conditional expectation of the causal derivative  $\frac{1}{T_i+1}$  given  $X_i = j$  is:

$$\sum_{t=0}^{\infty} \frac{1}{t+1} e^{-\lambda_j} \frac{\lambda_j^t}{t!} = \sum_{t=0}^{\infty} e^{-\lambda_j} \frac{\lambda_j^t}{(t+1)!}. \quad (2)$$

Define  $s := t + 1$ . Then we can rewrite the above as:

$$\sum_{s=1}^{\infty} e^{-\lambda_j} \frac{\lambda_j^{s-1}}{s!} = \frac{e^{-\lambda_j}}{\lambda_j} \sum_{s=1}^{\infty} \frac{\lambda_j^s}{s!} \quad (3)$$

$$= \frac{(e^{\lambda_j} - 1)}{\lambda_j e^{\lambda_j}} \quad (4)$$

$$= \frac{1 - e^{-\lambda_j}}{\lambda_j}. \quad (5)$$

where the second-to-last equality uses the identity  $\sum_{s=0}^{\infty} \frac{x^s}{s!} = e^x$ .

### A.2 Derivation of $T_i^*$

Within a given stratum  $X_i = j$ ,  $h(X_i) = E[T_i | X_i] = \lambda_j$  by assumption. We want to derive the point  $T_i^*$  between  $T_i$  and  $\lambda_j$  at which we are evaluating the derivative  $f'(t) = 1/(t+1)$ . We will condition on  $X_i$  throughout.

By the mean value theorem, there is some  $T_i^*$  between  $T_i$  and  $\lambda_j$  for which:

$$f(T_i) - f(\lambda_j) = f'(T_i^*)(T_i - \lambda_j). \quad (6)$$

Plugging in the definition of  $f$  and its derivative, we write this as:

$$\log(T_i + 1) - \log(\lambda_j + 1) = \frac{1}{T_i^* + 1}(T_i - \lambda_j). \quad (7)$$

Solving for  $T_i^*$  yields:

$$T_i^* = \frac{T_i - \lambda_j}{\log \frac{T_i + 1}{\lambda_j + 1}} - 1. \quad (8)$$

Note that, when  $T_i = \lambda_j$ , this quantity is undefined (in which case we just set  $T_i^* = T_i$ ).

In our simulations, we estimate the theoretical plim of RORR by taking many draws of  $T_i$ , plugging them into this formula, and using the resulting stratum means to estimate the RORR plim.

## B Choosing the Number of Segments

Choosing the number of segments  $K$  into which to coarsen the treatment involves the usual considerations of bias and variance. Larger values of  $K$  reduce bias, but they also reduce the sample size in each segment, increasing variance. Increasing  $K$  can also make results less interpretable. In our day-to-day work, we find that a relatively small number of bins—for example, dividing users into low, medium, and high usage segments—is sufficient to uncover meaningful heterogeneity in the dose-response function.

Here, we give a highly informal justification for choosing a relatively small number of bins, on the order of  $N^{1/5}$ . In practice, researchers should assess the sensitivity of their results to different choices of  $K$ .

We assume the usual conditions for the coarsened AIPW estimator, i.e., that the treatment is conditionally unconfounded and is continuously distributed on a compact interval. As in the main text, we decompose the error

of the coarsened AIPW estimator as

$$\begin{aligned}
& \hat{\psi} - E[f'(T_i)] \\
&= \underbrace{\left\{ \sum_{k=1}^{K-1} w_k \left( \frac{\psi_{k+1} - \psi_k}{\ell} \right) - E[f'(T_i)] \right\}}_{\text{coarsening bias}} \\
&+ \underbrace{\sum_{k=1}^{K-1} w_k \left( \frac{(\hat{\psi}_{k+1} - \psi_{k+1}) - (\hat{\psi}_k - \psi_k)}{\ell} \right)}_{\text{estimation error}}. \tag{9}
\end{aligned}$$

Theorem 1 showed that the coarsening bias is

$$\sum_{k=1}^{K-1} w_k \left( \frac{\psi_{k+1} - \psi_k}{\ell} \right) - E[f'(T_i)] = O(\ell) = O(K^{-1}), \tag{10}$$

and therefore the squared coarsening bias is  $O(K^{-2})$ .

Next, we consider sampling variation. With cross-fitting and the usual AIPW regularity conditions, we have that

$$\hat{\psi}_k - \psi_k = \frac{1}{N} \sum_{i=1}^N \phi_k(O_i) + R_{k,N}, \tag{11}$$

where  $\phi_k$  is the AIPW influence function and  $R_{k,N}$  is a higher-order remainder due to nuisance estimation. Under smoothness of  $p(t \mid x)$ ,  $\pi_k = O(\ell)$ . Therefore, the variance of the bin-level AIPW-estimated mean is

$$\text{Var}(\hat{\psi}_k) = O\left(\frac{1}{N\ell}\right) = O\left(\frac{K}{N}\right). \tag{12}$$

It remains to account for the remainder due to nuisance estimation. Let

$$r_{N,K} = \max_{1 \leq k \leq K} |R_{k,N}|. \tag{13}$$

Plugging this into the estimation error expression (and ignoring telescoping for a conservative bound) yields

$$\left| \sum_{k=1}^{K-1} w_k \left( \frac{R_{k+1,N} - R_{k,N}}{\ell} \right) \right| = O_p(Kr_{N,K}), \tag{14}$$

so its squared contribution is  $O_p(K^2 r_{N,K}^2)$ .

Combining these pieces, the coarsened AIPW estimator has MSE

$$\text{MSE}(\hat{\psi}) = O(K^{-2}) + O\left(\frac{K}{N}\right) + O_p(K^2 r_{N,K}^2). \quad (15)$$

A typical assumption for the nuisance remainder is that it is  $o_p((N\ell)^{-1/2})$ , so a conservative bound sets

$$r_{N,K} = O_p\left(\left(\frac{N}{K}\right)^{-1/2}\right) = O_p\left(\sqrt{\frac{K}{N}}\right). \quad (16)$$

Under this assumption, the nuisance-remainder contribution is

$$O_p(K^2 r_{N,K}^2) = O_p\left(\frac{K^3}{N}\right), \quad (17)$$

and so

$$\text{MSE}(\hat{\Psi}_K) = O(K^{-2}) + O\left(\frac{K^3}{N}\right), \quad (18)$$

up to lower-order terms. The  $K$  that minimizes this is  $K^* = O(N^{1/5})$

Again, this derivation is heuristic and should be treated as a starting point. Researchers should assess the sensitivity of their results to several values of  $K$  while also scrutinizing the underlying assumptions of continuity and positivity of the treatment.

## C Lemma 1

**Lemma 1.** *Assume  $f$  is twice continuously differentiable with  $|f''(t)| \leq M$  and that  $p(t | x)$  is twice continuously differentiable with uniformly bounded derivatives. Also assume positivity:*

$$\int_{S_k} p(t | x) dt \geq \epsilon \ell$$

*uniformly in  $k$  and  $x$ .*

*Then for every  $x$  and uniformly in  $k$ ,*

$$\int f(t) r_k(t|x) dt = f(\bar{t}_k) + O(\ell^2), \quad (19)$$

*where, as in the main text,  $\bar{t}_k = \frac{t_{k+1} + t_k}{2}$ ,  $\ell = t_{k+1} - t_k$ , and  $r_k(t|x) = \frac{p(t|x)1(t \in S_k)}{\int_{S_k} p(t|x) dt}$ .*

*Proof.* Since  $f$  is twice continuously differentiable,

$$\begin{aligned} \int f(t)r_k(t|x)dt &= f(\bar{t}_k) + f'(\bar{t}_k) \int (t - \bar{t}_k)r_k(t|x)dt \\ &\quad + \frac{1}{2} \int f''(\tilde{t})(t - \bar{t}_k)^2 r_k(t|x)dt \end{aligned} \quad (20)$$

for some  $\tilde{t}$  between  $t$  and  $\bar{t}_k$ .

Since  $f''$  is bounded and  $(t - \bar{t}_k)^2 \leq (\ell/2)^2$ ,

$$\left| \frac{1}{2} \int f''(\tilde{t})(t - \bar{t}_k)^2 r_k(t|x)dt \right| \leq C\ell^2. \quad (21)$$

Therefore, it remains to show that the other error term  $\int (t - \bar{t}_k)r_k(t|x)dt = O(\ell^2)$ .

We can rewrite the numerator of this term as

$$\int_{-\ell/2}^{\ell/2} up(\bar{t}_k + u|x)du, \quad (22)$$

where  $u := t - \bar{t}_k$ . Since  $p$  is twice continuously differentiable, this is equal to

$$\int_{-\ell/2}^{\ell/2} u \left[ p(\bar{t}_k|x) + up'(\bar{t}_k|x) + \frac{1}{2}u^2p''(\xi|x) \right] du \quad (23)$$

for some  $\xi$  between  $t$  and  $\bar{t}_k$ . The first term integrates to zero because  $\int_{-\ell/2}^{\ell/2} udu = 0$ . Therefore, the numerator depends only on second- and higher-order powers of  $u$ . Since  $u^2 \leq \ell^2$ , the numerator is  $O(\ell^3)$ .

Turning to the denominator and defining  $u$  as before, we have

$$\int_{S_k} p(t|x)dt = \int_{-\ell/2}^{\ell/2} p(\bar{t}_k + u|x)du. \quad (24)$$

By the same expansion, this is  $p(\bar{t}_k|x)\ell + O(\ell^2)$ . Putting the numerator and denominator together, we have  $O(\ell^3)/(p(\bar{t}_k|x)\ell + O(\ell^2)) = O(\ell^2)$  by positivity, which gives the result.  $\square$